

Using Empathic Mixed Reality Agents for Remote Collaboration

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ABSTRACT

This paper describes research on using Empathic Mixed Reality Agents (EMiRAs) for enhancing remote collaboration. Adopting Mixed Reality (MR) for remote collaboration is a promising solution which uses virtual avatars and visual cues to allow people to communicate naturally in three-dimensional (3D) MR environments. Advances in Artificial Intelligence (AI) make it possible to form hybrid teams of humans and AI driven virtual agents in MR environments. Forming hybrid teams to enhance human-to-human collaboration could be a promising way to take advantage of the complementary nature of human and AI agents. However, there still needs to be research conducted on how hybrid teams like this can improve MR remote collaboration. My research is focused on designing and evaluating EMiRAs for enhancing human-to-human remote collaboration. In this doctoral consortium presentation, I will introduce the EMiRA concept by identifying the convergence of Empathic Computing, MR and AI Agent. In the paper, I discuss the research motivation, recently published study results and plans for future work.

Index Terms: Human-centered computing—Human computer interaction (HCI)—empirical study; Human-centered computing—Collaborative and social computing—Empirical studies in collaborative and social computing

1 INTRODUCTION

Advances in Artificial Intelligence (AI) and Mixed Reality (MR) have now allowed human-like Intelligent Virtual Agents (IVAs) to move from being shown on two-dimensional (2D) screens to being experienced in three-dimensional (3D) real and virtual worlds [5], enabling natural face-to-face interaction. Mixed Reality Agents (MiRAs) [3] are Agents embodied in 3D MR environments, which can occupy and interact with both virtual and real environments. Human-like IVAs embodied in MR are a kind of MiRAs with human-like characteristics, such as communicating through natural language, expressing human-like behaviors and even emotions. Such agents that can show empathy for others and by their design elicit human empathy are called empathic agents [6]. These could be promising for providing mental health treatment, online education, daily assistance and in other applications. For example, Rizzo et al. [8] used a virtual human to play the role of clinical interviewer to reduce the stigma of seeking mental health treatment. Virtual humans like this could also be enhanced to understand and respond to the user's state by sensing physiological and behavioral signals such as brain waves, facial expressions and voices. To create deeper empathy between human and IVAs, both empathic agents and MiRAs rely on technologies like implicit understanding, experience capture, and natural communication, and Empathic Computing [1].

Motivated by the concepts of MR, IVAs and Empathic Computing, I am conducting research into interactions with empathic agents embodied in MR environments, in an area called Empathic Mixed

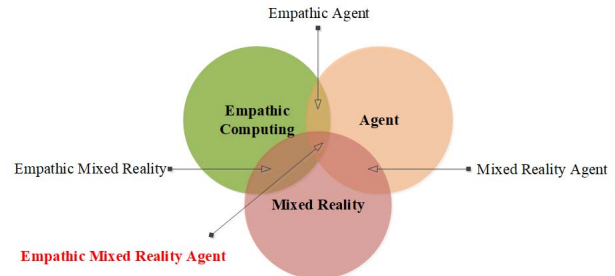


Figure 1: The Empathic Mixed Reality Agents (EMiRA) venn diagram.

Reality Agents (EMiRAs). As shown in Fig. 1, EMiRAs combine empathic agents, MiRA and Empathic Mixed Reality (EMR) [7]. EMR is a kind of MR system that adopts Empathic Computing technologies to create deeper understanding or empathy between people, especially in MR remote collaboration. Previously researchers have explored hybrid teams [10, 11] consisting of real humans, AI-enabled agents and virtual avatars of remote collaborators. However, there has been little research on the benefits of adopting EMiRAs in hybrid teams. Therefore, in my doctoral research, I propose to design and evaluate EMiRAs for remote collaboration. I mainly focus on using physiological sensors to enable the EMiRAs's intelligent capabilities of understanding users implicitly.

2 RESEARCH GOALS AND QUESTIONS

My research explores designing and evaluating EMiRAs for human-to-human remote collaboration. The overall research hypothesis is that EMiRA could be used to enhance human-to-human remote collaboration. This ultimate research focus can be decomposed into two sub-objectives, **1)** investigating the factors that influence the design of EMiRAs for effective human-agent interaction, and **2)** examining whether EMiRAs designed using the factors explored in Objective **1** can effectively mediate MR remote collaboration.

Based on these research objectives, I will explore the following research questions to identify the factors that are likely to influence the designing of EMiRAs and explore the possible benefits of adopting EMiRAs for remote collaboration:

- **RQ1:** How do multimodal communication cues of EMiRAs influence human perception and behaviours?
- **RQ2:** How do the EMiRAs capabilities of sensing and interacting with mixed reality environments impact human empathy for the agents?
- **RQ3:** How can EMiRAs effectively mediate human-to-human remote collaboration?

RQ1 and **RQ2** focus on the basic traits and intelligent capabilities of an EMiRA, while the **RQ3** is about exploring the use of such agents for remote collaboration and evaluating the effectiveness of such agents in mediating human-to-human remote collaboration.

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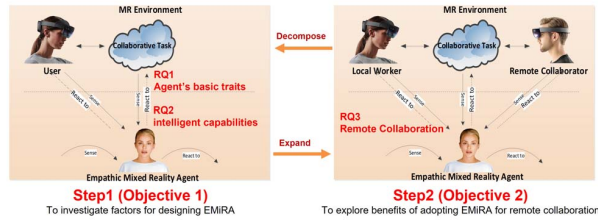


Figure 2: Relationships between the research steps and questions.

3 RESEARCH METHODS AND PROGRESS

Using a two step process, I plan to use both qualitative and quantitative Human Computer Interaction (HCI) research methods [4] to address the three research questions. As shown in Fig. 2, **RQ1** and **RQ2** will be addressed in step one and the **RQ3** will be researched in step two.

Firstly, to address **RQ1** I have conducted a pilot study [2] which explored the impact of virtual agents' multimodal communication on user brain activity and cognitive load. I developed a desert survival game where the participants made decisions on ranking 15 items collaboratively with the virtual agent in VR to achieve the highest survival probability possible. When asked, the virtual agent made up to ten suggestions to help the user. I evaluated three different communication methods based on a within-subject design: 1) a Voice-only Agent, 2) an Embodied Agent with speech and gaze, and 3) a Gestural Agent pointing at the objects while talking about them.

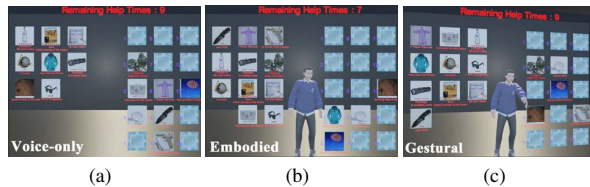


Figure 3: Three agent types: (a) the Voice-only Agent, (b) the Embodied Agent, (c) the Gestural Agent.

We captured participants' brain activities during the task using Electroencephalogram (EEG) to measure cognitive load and attention demands. We also used post-study questionnaires to measure subjective feelings of workload and copresence. The subjective results indicated that the Gestural agent reduced task load and enhanced copresence. The EEG results showed that providing a virtual body for the sudden speech could avoid an abrupt attention demand when the agent started speaking. Moreover, a sudden gesture coupled with the speech induced more attention demands, even though the speech was matched with the virtual body. This work was the first to explore the impact of IVAs' interaction methods in VR on users' brain activity, and our findings help inform IVAs interaction design.

Secondly, to answer **RQ2**, I plan to run a user study to evaluate the effect of EMiRA's capabilities of understanding the user's state by capturing their physiological signals (e.g., EEG, Galvanic Skin Response (GSR) and heart rate) which interacting with such agents. Similar to the pilot study above, I will use VR as interaction environment to suppress interference from real world. The Octopus sensing library [9] will be used to combine multiple physiological cues together and calculate the users arousal and valence emotional measures. After this study, I will further explore the impact of EMiRA's capabilities of responding to real world events in MR envi-

ronment on user experience. For example, designing an EMiRA that could sense environment temperature to adjust empathic response towards users.

Thirdly, to answer **RQ3**, I will mainly focus on adopting an EMiRA to form hybrid teams with multiple human collaborators. For example, using an EMiRA to work collaboratively with two people on desert survival game as we used in the pilot study presented above. One human is working in MR environment while another is in VR environment. The EMiRA could discuss and even debate with both human partners. This EMiRA will be designed based on the results from answering **RQ1** and **RQ2**. A user study will be conducted to explore benefits of using such an EMiRA for human-to-human remote collaboration in MR environment.

4 CONCLUSION

In this paper, I introduce the concept of Empathic Mixed Reality Agents (EMiRAs) by examining the intersection of research in Mixed Reality, Empathic Computing and Intelligent Agents. Inspired by the idea of forming hybrid teams consisting of real human and algorithm driven agents to make full use of both human and the agents, I am particularly interested in designing and evaluating EMiRAs for enhancing human-to-human remote collaboration. Based on this general objective, I propose three research questions and plan to adopt two research steps to handle the agent design and evaluation problems. A pilot study has already been completed to address the first research question, finding that using expressive virtual characters can reduce task load and enhance co-presence. In the next step of my thesis research, I will focus on the EMiRA's capabilities of sensing and interacting with the MR environment and evaluate how such intelligence could improve collaborative user experiences.

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